

The Technology Value Stream

Lead & Process Times, Deployment Flow, and DevOps

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Agenda



Technology Value Stream



Lead vs. Process Time



Common Scenario (Months)



DevOps Ideal (Minutes)



Comparison & Conclusions

What is the Technology Value Stream?

A business idea is transformed into a technology-enabled solution that adds value through the technology value stream. Product management, development, quality assurance, IT operations, and security are all areas where Lean principles are applied.



Lead Time vs. Process Time

Lead Time (LT) is the total time elapsed from start to finish, including wait, inspection, move, and queuing periods.

The actual amount of time spent working on a unit is known as Process Time (PT). $LT = PT$ plus idle time: Value streams are frequently dominated by waiting.



Common Scenario: Months-Long Deployments

Tightly connected, monolithic apps that impede change. Testing by hand, using a lot of resources, and having few test environments. Delays in provisioning and cutover during infrastructure deployment. Heroic efforts are required to deliver and resolve issues.



Complexity



Manual QA



Provisioning



Delay

DevOps Ideal: Minutes-Long Deployments

Iteration is made possible by quick, continuous feedback loops. Atomic, tiny code modifications lower risk and facilitate recovery. Automated testing enables QA to focus on complex issues. Teams can deploy independently thanks to modular designs.



Speed



Confidence



Automation

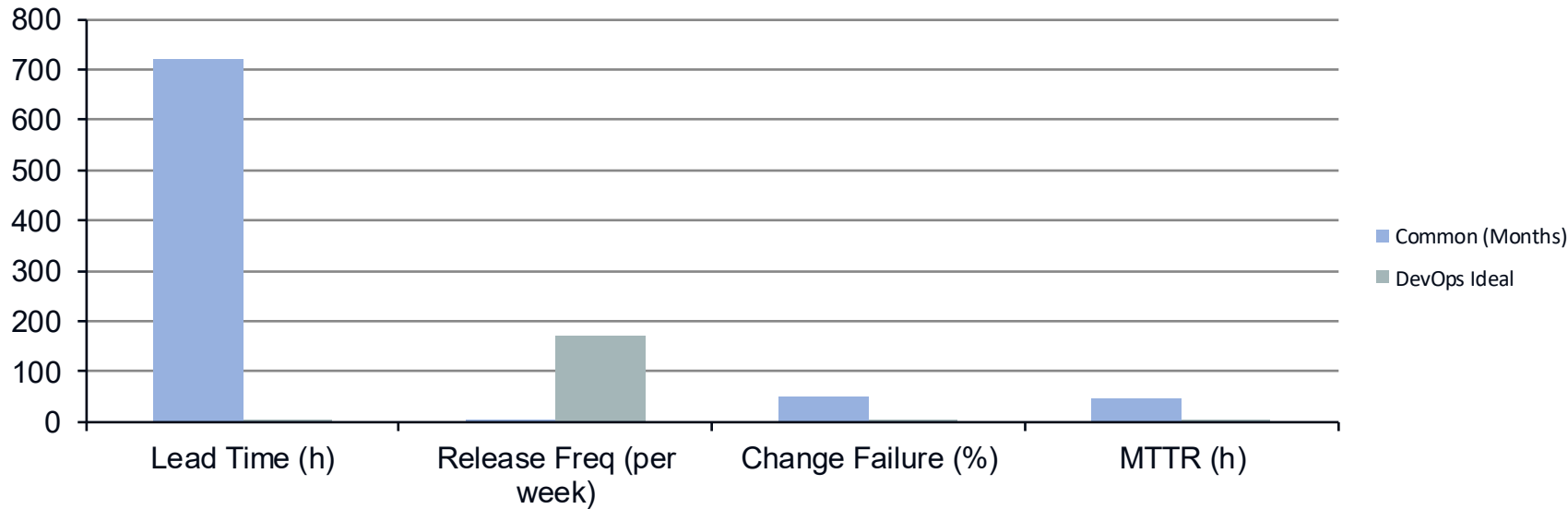


Modularity

"As an organisation goes from deployment lead times measured in months to minutes, the constraint moves in predictable ways."

– Gene Kim

Common vs. Ideal: Quantitative Contrast



Conclusions & Next Steps

- Understand your technology value stream: map it end-to-end and identify waste.
- Reduce lead time by attacking idle time: automate manual steps, eliminate bottlenecks.
- Break down monoliths into modular, decoupled services and deploy small changes often.
- Foster a culture of collaboration, continuous learning and fast feedback.

Ready to optimise your value stream?

Start your journey today!

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